Machine Learning

(Course Code: 18B1WCI634)

Multiple Linear Regression

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Multiple Linear Regression (MLR)

Definition:

MLR is an extension of simple linear regression used to predict a dependent variable (outcome) using two or more independent variables.

Equation:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \epsilon$$

y: Dependent variable (target)

β_0: Intercept

β_i: Coefficients (slopes) showing effect of each predictor

x_i: Independent variables (features)

ϵ: Error term (unexplained variability)

Assumptions of Multiple Linear Regression

Linearity: The relationship between predictors and the response is a straight line.

Independence: Each observation is independent of the others.

Homoscedasticity: The errors ((differences between actual and predicted values))have constant spread (same variance).

Normality: The errors follow a bell-shaped (normal) distribution.

No Multicollinearity: Predictors are not strongly related to each other.

General MLR Model

For two predictors x_1 and x_2, the regression model is:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$$

y: dependent variable

x_1, x_2: independent variables

β_0: intercept

β_1, β_2: regression coefficients

ϵ: error term

Objective Function (Least Squares)

We estimate coefficients by minimizing Sum of Squared Errors (SSE):

$$S(\beta_0, \beta_1, \beta_2) = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2})^2$$

Normal Equations

Step 1: Take partial derivatives

$$\frac{\partial S}{\partial \beta_0} = -2\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2}) = 0$$

$$\frac{\partial S}{\partial \beta_1} = -2\sum_{i=1}^{n} x_{i1}(y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2}) = 0$$

$$\frac{\partial S}{\partial \beta_2} = -2\sum_{i=1}^{n} x_{i2}(y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2}) = 0$$

Expansion of First Normal Equation

$$\frac{\partial S}{\partial \beta_0} = -2\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2}) = 0$$

Step 1: Expand the summation

$$\sum y_i - \sum \beta_0 - \sum \beta_1 x_{i1} - \sum \beta_2 x_{i2} = 0$$

Step 2: Simplify terms

$$\sum \beta_0 = n\beta_0, \quad \sum \beta_1 x_{i1} = \beta_1 \sum x_{i1}, \quad \sum \beta_2 x_{i2} = \beta_2 \sum x_{i2}$$

Final Form:

$$\sum y_i = n\beta_0 + \beta_1 \sum x_{i1} + \beta_2 \sum x_{i2}$$

Expansion of Second Normal Equation

$$\frac{\partial S}{\partial \beta_1} = -2 \sum_{i=1}^{n} x_{i1} \left(y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2} \right) = 0$$

Step 1: Expand the summation

$$\sum x_{i1} y_i - \sum x_{i1} \beta_0 - \sum \beta_1 x_{i1}^2 - \sum \beta_2 x_{i1} x_{i2} = 0$$

Step 2: Simplify terms

$$\sum x_{i1} \beta_0 = \beta_0 \sum x_{i1}, \quad \sum \beta_1 x_{i1}^2 = \beta_1 \sum x_{i1}^2, \quad \sum \beta_2 x_{i1} x_{i2} = \beta_2 \sum x_{i1} x_{i2}$$

Final Form:

$$\sum x_{i1} y_i = \beta_0 \sum x_{i1} + \beta_1 \sum x_{i1}^2 + \beta_2 \sum x_{i1} x_{i2}$$

Expansion of Third Normal Equation

From

$$\frac{\partial S}{\partial \beta_2} = -2 \sum_{i=1}^{n} x_{i2} \left(y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2} \right) = 0$$

Step 1: Expand the summation

$$\sum x_{i2} y_i - \sum x_{i2} \beta_0 - \sum \beta_1 x_{i1} x_{i2} - \sum \beta_2 x_{i2}^2 = 0$$

Step 2: Simplify terms

$$\sum x_{i2} \beta_0 = \beta_0 \sum x_{i2}, \quad \sum \beta_1 x_{i1} x_{i2} = \beta_1 \sum x_{i1} x_{i2}, \quad \sum \beta_2 x_{i2}^2 = \beta_2 \sum x_{i2}^2$$

Final Form:

$$\sum x_{i2} y_i = \beta_0 \sum x_{i2} + \beta_1 \sum x_{i1} x_{i2} + \beta_2 \sum x_{i2}^2$$

Expanded Normal Equations

$$\sum y_i = n\beta_0 + \beta_1 \sum x_{i1} + \beta_2 \sum x_{i2}$$

$$\sum x_{i1} y_i = \beta_0 \sum x_{i1} + \beta_1 \sum x_{i1}^2 + \beta_2 \sum x_{i1} x_{i2}$$

$$\sum x_{i2} y_i = \beta_0 \sum x_{i2} + \beta_1 \sum x_{i1} x_{i2} + \beta_2 \sum x_{i2}^2$$

From Expanded to Matrix Form

Step 1: Stack equations into matrix form

$$\begin{bmatrix} \sum y_i \\ \sum x_{i1} y_i \\ \sum x_{i2} y_i \end{bmatrix} = \begin{bmatrix} n & \sum x_{i1} & \sum x_{i2} \\ \sum x_{i1} & \sum x_{i1}^2 & \sum x_{i1} x_{i2} \\ \sum x_{i2} & \sum x_{i1} x_{i2} & \sum x_{i2}^2 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$$

Step 2: Compact representation

$$X^T X \beta = X^T y$$

Computation of $X^T X$ and $X^T y$

Design matrix:

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} \\ 1 & x_{21} & x_{22} \\ \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$X^T X = \begin{bmatrix} n & \sum x_{i1} & \sum x_{i2} \\ \sum x_{i1} & \sum x_{i1}^2 & \sum x_{i1} x_{i2} \\ \sum x_{i2} & \sum x_{i1} x_{i2} & \sum x_{i2}^2 \end{bmatrix}, \quad X^T y = \begin{bmatrix} \sum y_i \\ \sum x_{i1} y_i \\ \sum x_{i2} y_i \end{bmatrix}$$

Closed Form Solution

Closed form solution:

$$\beta = (X^T X)^{-1} X^T y$$

This yields estimates for

$$\beta = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix}$$

Numerical Example: MLR with Two Predictors

Data (n = 3):

i	x_{i1}	x_{i2}	y_i
1	1	2	6
2	2	0	8
3	3	1	13

Form X, $X^T X$ and $X^T y$

Design matrix and response:

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} \\ 1 & x_{21} & x_{22} \\ 1 & x_{31} & x_{32} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 0 \\ 1 & 3 & 1 \end{bmatrix}, \quad y = \begin{bmatrix} 6 \\ 8 \\ 13 \end{bmatrix}$$

Then

$$X^T X = \begin{bmatrix} 3 & 6 & 3 \\ 6 & 14 & 5 \\ 3 & 5 & 5 \end{bmatrix}, \quad X^T y = \begin{bmatrix} 27 \\ 61 \\ 25 \end{bmatrix}.$$

Inverse and Coefficient Estimates

Compute inverse of $X^T X$ (exact fractions shown):

$$(X^T X)^{-1} = \begin{bmatrix} 5 & -\frac{5}{3} & -\frac{4}{3} \\ -\frac{5}{3} & \frac{2}{3} & \frac{1}{3} \\ -\frac{4}{3} & \frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

Now the closed-form solution:

$$\hat{\beta} = (X^T X)^{-1} X^T y = \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix}.$$

So $\hat{\beta}_0 = 0,\ \hat{\beta}_1 = 4,\ \hat{\beta}_2 = 1.$

Model and Quick Check

Estimated regression model:

$$\hat{y} = 0 + 4x_1 + 1x_2.$$

Prediction Example

Predictions for the three training observations:

$$\begin{aligned} \hat{y}_1 &= 4(1) + 1(2) = 6 \quad \text{(actual 6)} \\ \hat{y}_2 &= 4(2) + 1(0) = 8 \quad \text{(actual 8)} \\ \hat{y}_3 &= 4(3) + 1(1) = 13 \quad \text{(actual 13)} \end{aligned} \tag{1}$$

Residuals are zero for this toy example (perfect fit).

Prediction for a new observation ($x_1 = 5, x_2 = 6$) :

$$\hat{y}_{\text{new}} = 4(5) + 1(6) = 26 \tag{2}$$

Assignment-1

A small company wants to model monthly sales (in thousands of units) using advertising spend on **TV** (\$1000s) and **Online** (\$1000s).
The observed data for three months are:

$$(\text{TV}_i,\ \text{Online}_i,\ \text{Sales}_i) \in \{(1,2,6),\ (2,1,5),\ (3,3,10)\}$$

(a) Write the multiple linear regression model

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i,$$

where $x_{i1} = \text{TV}_i$, $x_{i2} = \text{Online}_i$,

(b) estimate the regression coefficients $\beta_0, \beta_1, \beta_2$ and predict y when $x_1 = 4, x_2 = 2$.

Numerical Solution: Data and Sums

Data:

i	$x_{i1} = \text{TV}_i$	$x_{i2} = \text{Online}_i$	$y_i = \text{Sales}_i$
1	1	2	6
2	2	1	5
3	3	3	10

Form X, $X^T X$ and $X^T y$

Design matrix and response vector:

$$X = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 1 & 3 & 3 \end{bmatrix}, \qquad y = \begin{bmatrix} 6 \\ 5 \\ 10 \end{bmatrix}.$$

Then

$$X^T X = \begin{bmatrix} 3 & 6 & 6 \\ 6 & 14 & 13 \\ 6 & 13 & 14 \end{bmatrix}, \qquad X^T y = \begin{bmatrix} 21 \\ 46 \\ 47 \end{bmatrix}.$$

Compute $(X^T X)^{-1}$ For this matrix:

$$(X^T X)^{-1} = \begin{bmatrix} 3 & -\frac{2}{3} & -\frac{2}{3} \\ -\frac{2}{3} & \frac{2}{3} & -\frac{1}{3} \\ -\frac{2}{3} & -\frac{1}{3} & \frac{2}{3} \end{bmatrix}.$$

Now compute the closed-form estimate:

$$\hat{\beta} = (X^T X)^{-1} X^T y = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}.$$

So $\hat{\beta}_0 = 1,\ \hat{\beta}_1 = 1,\ \hat{\beta}_2 = 2.$

Estimated Model and Training Check

Estimated regression equation (units: sales in thousands of units per month, ad spends in \$1000s):

$$\boxed{\hat{y} = 1 + 1 \cdot x_1 + 2 \cdot x_2}$$